2.34 Mean Value – Part 2

Worked Solutions

1a)

The mean value on $[a, b]$ is $\frac{1}{b-a}\int_a^b f(x)\,dx$. Thus

$$\frac{1}{\sqrt{2}/2}\int_0^{\sqrt{2}/2} \pi x\cos(\pi x^2)\,dx = \frac{1}{\sqrt{2}/2}\left[\frac{1}{2}\sin(\pi x^2)\right]_0^{\sqrt{2}/2} = \frac{1}{\sqrt{2}}.$$

1b)

$$\frac{1}{\pi/3-\pi/4}\int_{\pi/4}^{\pi/3} \operatorname{cosec}^2 x\,dx = \frac{12}{\pi}\left[-\cot x\right]_{\pi/4}^{\pi/3} = \frac{12-4\sqrt{3}}{\pi}.$$

1c)

$$\int_0^1 \frac{x^2}{5x^3+1}\,dx = \left[\frac{1}{15}\ln|5x^3+1|\right]_0^1 = \frac{1}{15}\ln 6.$$

The interval has length 1, so the mean value is $\frac{1}{15}\ln 6$.

1d)

Using $\sin^2 A = \frac{1}{2}(1-\cos 2A)$,

$$\begin{aligned}\frac{4}{\pi}\int_0^{\pi/4} \sin^2 4x\,dx &= \frac{4}{\pi}\int_0^{\pi/4}\frac{1}{2}(1-\cos 8x)\,dx\\ &= \frac{4}{\pi}\left[\frac{x}{2}-\frac{\sin 8x}{16}\right]_0^{\pi/4} = \frac{1}{2}.\end{aligned}$$

1e)

$$\frac{1}{3}\int_2^5 2^x\,dx = \frac{1}{3\ln 2}\left[2^x\right]_2^5 = \frac{28}{3\ln 2}.$$

1f)

$$\frac{12}{\pi}\int_0^{\pi/12} \sin 2x\cos^3 2x\,dx = \frac{12}{\pi}\left[-\frac{1}{8}\cos^4 2x\right]_0^{\pi/12} = \frac{21}{32\pi}.$$

2)

The worked solution uses $f(x) = xe^{-3x}$. Integration by parts gives

$$\int xe^{-3x}\,dx = -\frac{x}{3}e^{-3x} - \frac{1}{9}e^{-3x} + c.$$

Therefore the mean value as worked in the source solutions is

$$\frac{1}{a}\int_0^a xe^{-3x}\,dx = \frac{1-(3a+1)e^{-3a}}{9a}.$$

3)

$$\frac{1}{8}\int_0^8 \left(\frac{2}{\sqrt{3x+1}}+k\right)dx = \frac{1}{8}\left[\frac{4}{3}\sqrt{3x+1}+kx\right]_0^8 = \frac{2}{3}+k.$$

Since this mean value is 2, $k=\frac{4}{3}$.

4)

The mean value of f is

$$\frac{1}{3}\int_0^3 \sqrt{x+1}\,dx = \frac{1}{3}\left[\frac{2}{3}(x+1)^{3/2}\right]_0^3 = \frac{14}{9}.$$

The mean value of g on $[0,k]$ is

$$\frac{1}{k}\int_0^k \frac{3}{7}x^2\,dx = \frac{k^2}{7}.$$

Thus $\frac{14}{9}=\frac{1}{2}\cdot\frac{k^2}{7}$, giving $k^2=\frac{196}{9}$. Since k is positive, $k=\frac{14}{3}$.

5a)

$$\frac{1}{20}\int_0^{20}\frac{30t}{t+5}\,dt = \frac{1}{20}\left[30t-150\ln|t+5|\right]_0^{20} = 17.9 \text{ m s}^{-1} \quad (3 \text{ s.f.}).$$

5b)

$$a=\frac{dv}{dt}=\frac{150}{(t+5)^2}.$$

Therefore the mean acceleration on the interval is

$$\frac{1}{20}\int_0^{20}\frac{150}{(t+5)^2}\,dt = \frac{15}{2}\left[-(t+5)^{-1}\right]_0^{20} = 1.2 \text{ m s}^{-2}.$$

6a)

$$\frac{1}{3}\int_1^4 \ln x\,dx = \frac{1}{3}[x\ln x - x]_1^4 = \frac{4}{3}\ln 4 - 1.$$

6b)

Adding a constant c to a function adds c to its mean value. Hence

$$\frac{4}{3}\ln 4 - 1 + c = \frac{4}{3}\ln 4,$$

so $c=1$.

7)

The signed area between $y=\sin x$ and $y=k$ from $x=0$ to $x=\pi/2$ is zero because the two shaded areas are equal. Thus

$$\int_0^{\pi/2}(\sin x - k)\,dx = 0, \qquad 1-\frac{k\pi}{2}=0,$$

so $k=\frac{2}{\pi}$.

8)

$$\mathrm{RMS}(f) = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} 16\cos^2 x\,dx}.$$

Using $\cos^2 x = \frac{1}{2}(1+\cos 2x)$,

$$\int_0^{2\pi} 16\cos^2 x\,dx = 8\left[x + \frac{1}{2}\sin 2x\right]_0^{2\pi} = 16\pi.$$

Therefore $\mathrm{RMS}(f) = \sqrt{8} = 2\sqrt{2}$.

9a)

Using $\cos^2 A = \frac{1}{2}(1+\cos 2A)$,

$$\begin{aligned}\cos^4\theta &= \frac{1}{4}(1 + 2\cos 2\theta + \cos^2 2\theta)\\ &= \frac{1}{4}\left(1 + 2\cos 2\theta + \frac{1}{2}(1+\cos 4\theta)\right)\\ &= \frac{1}{8}(3 + 4\cos 2\theta + \cos 4\theta).\end{aligned}$$

9b)

Here $v^2 = a^2\cos^4(kt)$. The mean value of v^2 over one cycle $[0, \pi/k]$ is

$$\begin{aligned}\frac{k}{\pi}\int_0^{\pi/k} a^2\cos^4(kt)\,dt &= \frac{a^2 k}{8\pi}\int_0^{\pi/k}(3 + 4\cos 2kt + \cos 4kt)\,dt\\ &= \frac{3a^2}{8}.\end{aligned}$$

The effective speed is therefore $\sqrt{3a^2/8} = \frac{\sqrt{3}a}{2\sqrt{2}}$ m s^{-1}.

10a)

Since $\sinh x$ is an odd function and n is odd, $(\sinh x)^n$ is odd. Its integral over the symmetric interval $[-k\pi, k\pi]$ is zero, so its mean value is 0.

10b)

$$\cosh(ix) = \frac{e^{ix} + e^{-ix}}{2} = \cos x.$$

As n is odd, $(\cos x)^n$ is even, and its integral over $[-k\pi, k\pi]$ is $2\int_0^{k\pi}\cos^n x\,dx$. However, for odd n this integral is not generally zero over $[0, k\pi]$; evaluating it gives For odd n, $\cos^n x$ has opposite values at x and $\pi - x$, so its integral over each interval of length π is zero. Thus its mean value is 0.

10c)

$$\sinh(ix) = i\sin x.$$

For odd n, $(i\sin x)^n = i^n \sin^n x$ is an odd function of x. Its integral over $[-k\pi, k\pi]$ is zero, so its mean value is 0.

11a)

$$\frac{1}{3}\int_0^3 \frac{1}{9+x^2}\,dx = \frac{1}{3}\left[\frac{1}{3}\arctan\left(\frac{x}{3}\right)\right]_0^3 = \frac{\pi}{36}.$$

11b)

$$4\int_0^{1/4} \frac{1}{\sqrt{1-4x^2}}\,dx = 4\left[\frac{1}{2}\arcsin(2x)\right]_0^{1/4} = \frac{\pi}{3}.$$

11c)

$$\frac{1}{a}\int_0^a \frac{1}{x^2+7}\,dx = \frac{1}{a\sqrt{7}}\left[\arctan\left(\frac{x}{\sqrt{7}}\right)\right]_0^a = \frac{1}{a\sqrt{7}}\arctan\left(\frac{a}{\sqrt{7}}\right).$$

12)

Completing the square, $x^2 - 6x + 16 = 7 + (x-3)^2$. Hence

$$\begin{aligned}\frac{1}{6}\int_0^6 \frac{1}{\sqrt{x^2-6x+16}}\,dx &= \frac{1}{6}\left[\ln\left(x-3+\sqrt{7+(x-3)^2}\right)\right]_0^6 \\ &= \frac{1}{6}\ln 7.\end{aligned}$$

13)

$$\frac{1}{a-3}\int_3^a \frac{1}{\sqrt{x^2-9}}\,dx = \frac{1}{a-3}\left[\ln\left(x+\sqrt{x^2-9}\right)\right]_3^a = \frac{1}{a-3}\ln\left(\frac{a+\sqrt{a^2-9}}{3}\right).$$

14)

$$\frac{1}{4}\int_0^4 \left(12+8\,\mathrm{sech}^2(0.5t)\right)dt = \frac{1}{4}[12t+16\tanh(0.5t)]_0^4 = 12+4\tanh 2 = 15.9$$

to three significant figures, in watts per square metre.

15)

The second minute is the interval $1 \le t \le 2$:

$$\begin{aligned}\int_1^2 \left(50+20\,\mathrm{cosech}^2(0.2t)\right)dt &= [50t-100\coth(0.2t)]_1^2 \\ &= 50-100\coth(0.4)+100\coth(0.2) \\ &\approx 293 \text{ lumens.}\end{aligned}$$

The interval has length 1, so this is the mean brightness to three significant figures.

16a)

$$\frac{1}{12}\int_0^{12} \frac{18}{\sqrt{t^2+9}}\,dt = \frac{18}{12}\left[\ln\left(t+\sqrt{t^2+9}\right)\right]_0^{12} = \frac{3}{2}\ln(4+\sqrt{17}) = 3.14 \text{ m s}^{-1}$$

to three significant figures.

16b)

The mean acceleration is the change in velocity divided by the time interval:

$$\frac{v(12) - v(0)}{12} = \frac{6/\sqrt{17} - 6}{12} \approx -0.379 \text{ m s}^{-2}.$$

Thus the mean deceleration is 0.379 m s^{-2}.

17)

$$\frac{1}{a}\int_0^a \frac{1}{\sqrt{25 - x^2}}\,dx = \frac{1}{a}\left[\arcsin\left(\frac{x}{5}\right)\right]_0^a = \frac{1}{a}\arcsin\left(\frac{a}{5}\right).$$

Since $\cos\theta = a/5$, with $0 \le \theta \le \pi/2$, $\arcsin(a/5) = \pi/2 - \theta$. The mean value is therefore $\frac{\pi}{2a} - \frac{\theta}{a}$.

18)

Write

$$\frac{x+2}{(x^2+2)(x-1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}.$$

Equating coefficients gives $A = 1$, $B = -1$, and $C = 0$. Thus

$$f(x) = \frac{1}{x-1} - \frac{x}{x^2+2}.$$

The signed area from $x = 3/2$ to $x = 5/2$ is

$$\begin{aligned}\int_{3/2}^{5/2} f(x)\,dx &= \left[\ln|x-1| - \frac{1}{2}\ln(x^2+2)\right]_{3/2}^{5/2} \\ &= \ln\left(\frac{\sqrt{561}}{11}\right).\end{aligned}$$

The two areas are equal, so this signed area is zero after subtracting the rectangle of height k and width 1. Therefore

$$k = \ln\left(\frac{\sqrt{561}}{11}\right).$$